EXP 13
08-10-26

IMPLEMENTATION OF A PING PROGRAM.

AIM:-

To write a java program to ping a hostname or IP address using the Inet address class and check whether the host is reachable, along with the response time.

PROCEDURE:-

i) Open a java IDE or terminal and create a file named MyPing.java.

ii) Import InetAddress and Scanner classes.

iii) Get the hostname or IP address from the user.

iv) Use InetAddress.getByName() to obtain the IP address.

v) Use isReachable(3000) to check whether the host is reachable within 3 seconds.

vi) Calculate and display the response time in milliseconds.

vii) Repeat the ping 4 times with a 1-second delay between each attempt.

viii) Display "Reply from..." if the host is reachable; otherwise, display "Request timed out".

ix) Compile and execute the program.

RESULT:-

The Java program was successfully executed to ping the given hostname / IP address. It displays whether the host is reachable and choose the response time.

OUTPUT:-

```
Users\Luqman\Downloads\CN> javac PingProgram.java

Users\Luqman\Downloads\CN> java PingProgram

Enter host (e.g., google.com) : google.com

Pinging 142.250.206.78

Reply from 142.250.206.78 : time=80ms

Reply from 142.250.206.78 : time=1885ms

Reply from 142.250.206.78 : time=1019ms

Reply from 142.250.206.78 : time=73ms.
```

13- Implement a ping program.

CODE:

```
import socket
import struct
import time
import os.

def checkSum(data):
    sum=0
    for i in range(0, len(data)-1,2):
        sum += (data[1]<<8 + data[i+1]
    if len(data)%2:
        sum+= data[-1]<<8
    Sum = (sum >>16) + (sum 0xffff)
    Sum+ = Sum >>16.
    return ~Sum & 0ffff.

def ping(host):
 try:
    dest = socket.gethostbyname(host)
    sock = socket.socket(Socket.AF_INET; Socket.sock_RAW
                         Socket.IRPPTO_ICMP)
    socket.settimeout(1)
    packet_id =os.getpid() & 0xfff.
    sequence=1
    header = Struct.pack("!BBHHH",8,0,0, packet_id)
    data = struct.pack("id", time.sequence)
    checksum = checksum(header + data).
    header = struct.pack ("!BHHH", 8,0, checksum, packet_id,
                          sequence).
    packet= header+data.
```

```
while True:
    try:
        recieved address = sock.recvfrom(1024)
        end = time.time()
        break.
        print("Request timedout").
        return.
    delay = (end-start) * 1000.
    print("Reply From", address[0])
    print("Time = {:.2f}ms', format(delay)).
    Sock.(close().
```